Overview of Generative AI

Part 2: Diffusion Models + Day 1 Wrap-up

Day 1 - Session 4

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What We'll Cover

- How diffusion models work
- Why diffusion powers today's image generators
- Lab 1.4: generate images with a Hugging Face pipeline
- Day 1 recap and a look ahead to Day 2

Diffusion Models

Generating by removing noise

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The Core Idea

Diffusion models learn to turn noise into data, one small step at a time.

- Start from a field of pure random noise
- Remove a little noise, guided by what the model learned
- Repeat many times until a clean image appears

It is the reverse of gradually destroying an image with noise.

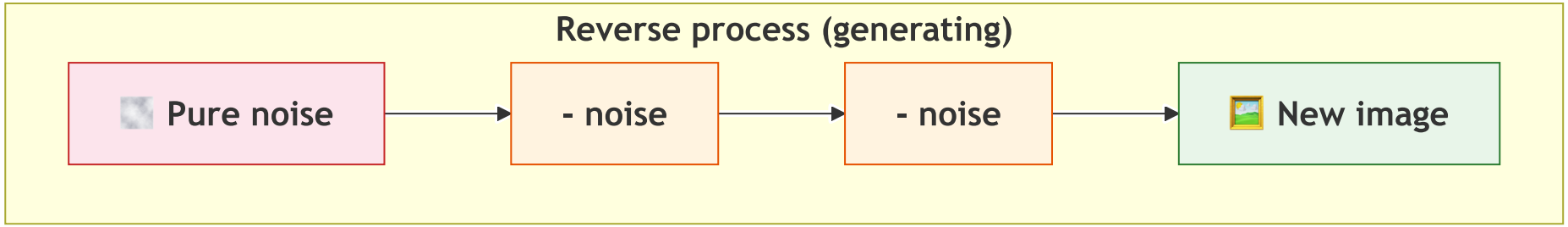
> **IMPORTANT**
>
> Diffusion never paints in one shot. It starts from noise and removes a little at a time, which is what keeps the output stable and sharp.

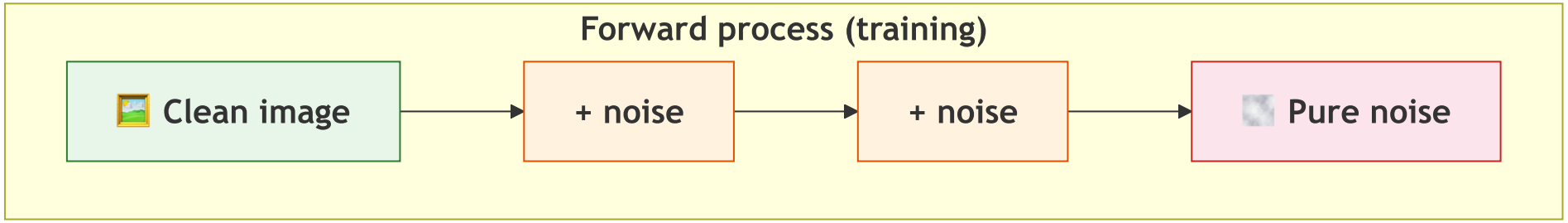
In Practice: Denoising

Imagine a sculptor carving a statue from a rough block of stone.

- **The Noise:** The rough block of stone has no clear shape.
- **The Process:** The sculptor chips away tiny pieces of stone, step by step, guided by a mental image of the subject.
- **The Result:** After many small adjustments, a detailed figure emerges. Diffusion does this with pixels instead of stone.

Forward and Reverse





Training adds noise step by step. Generating runs the learned denoiser backward.

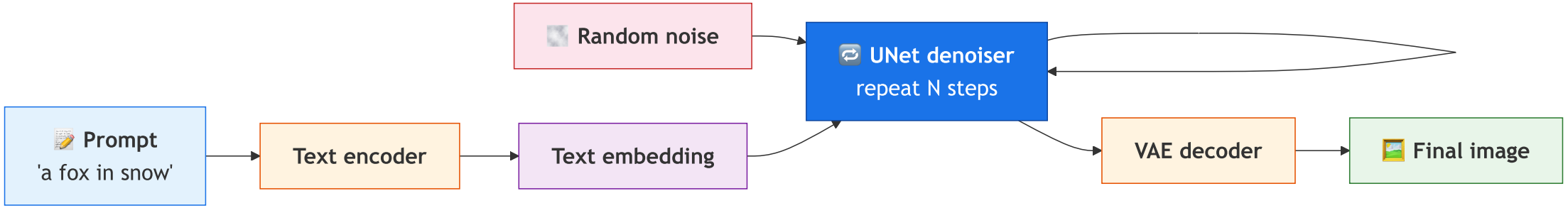
Why Train by Adding Noise?

- Adding noise is easy and gives the model perfect practice pairs
- At each step the model learns to predict the noise to remove
- Many small denoising steps are easier to learn than one big jump

Breaking generation into small steps is what makes the output sharp and stable.

> **TIP**
>
> Step count is your main speed-versus-quality dial: more steps usually means higher quality but slower generation.

Text-to-Image Pipeline



A text encoder turns the prompt into an embedding that steers the denoising toward your description.

In Practice: Steering with Text

The text prompt acts as the blueprint for the denoising process.

- **Architecture:** A firm needs a concept for a sustainable office building.
- **The Prompt:** They type "glass office building with green rooftop gardens, sunny day."
- **The Pipeline:** The model uses this text blueprint to decide which "noise" to remove at each step. It slowly sculpts the random pixels until a building with a green roof appears.

Intuit Scenario: Mailchimp

Mailchimp puts text-to-image generation in the hands of small businesses.

- **The brief:** an owner types "summer sale, bright colors, product on a beach"
- **The pipeline:** a diffusion model turns that prompt into several on-brand images
- **The payoff:** the owner picks one, adjusts the copy, and launches the campaign in minutes, with no designer needed

Diffusion vs VAE

	VAE	Diffusion
How it generates	Decode one latent sample	Denoise over many steps
Speed	Fast, one pass	Slower, many steps
Image quality	Softer	Sharp, detailed
Powers	Teaching, representations	Modern image generators

Where You See Diffusion

- Text-to-image tools (Stable Diffusion, DALL-E, Midjourney)
- Image editing: inpainting, outpainting, upscaling
- Growing use in audio and video generation

The same denoising idea carries across all of these.

Breakout Lab 1.4

Generate images with a diffusion pipeline

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Lab 1.4: Diffusion Image Generation

Open `day1/lab1_4_diffusion/start.ipynb` in Colab.

- Load a small pre-trained diffusion pipeline from Hugging Face
- Generate images by denoising from random noise (inference only, no training)
- Change the number of denoising steps and compare quality
- Set a seed to make a result repeatable

> Stretch goal: run text-to-image with Stable Diffusion (a T4 GPU runtime is recommended).

Note on Runtime

- This lab runs inference on a CPU or the free T4 GPU runtime, no paid GPU required
- Keep image size and step count small so each run finishes in class
- The first run downloads the model, so it takes longer than later runs

WARNING

The first generation downloads model weights and will feel slow. Keep image size and step count small so later runs finish quickly in class.

Day 1 Wrap-up

What you built today

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What You Built Today

- [x] A trained classifier with real evaluation metrics
- [x] An image classifier in PyTorch
- [x] A VAE that generates new handwritten digits
- [x] Images generated by a diffusion pipeline

From learning rules on data to creating brand-new images.

The Thread Through Day 1

- Machine learning learns patterns from data
- Deep learning stacks neurons to learn features and predictions
- Generative models learn a distribution and sample from it
- VAEs and diffusion are two ways to draw new samples

Looking Ahead to Day 2

- Language as data: tokenization, embeddings, similarity search
- Pre-trained models: BERT vs GPT and the Transformer
- Prompting techniques and responsible prompting
- Tool use and building a simple agent

Key Takeaways

1. Diffusion generates by removing noise step by step
2. Training pairs come free by adding noise to real images
3. A text encoder steers diffusion toward your prompt
4. Diffusion trades speed for sharp, detailed images
5. You generated images from a pre-trained pipeline, no training needed

Questions?

That's a wrap on Day 1

Afternoon Kahoot next

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